

Installation Guide

1. Prerequisites

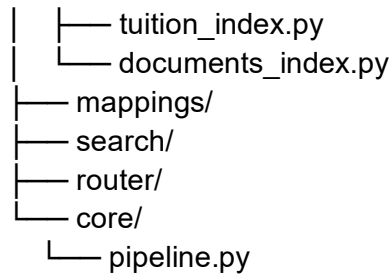
Before starting, ensure the following are installed on your macOS machine:

- macOS 12 or later
- Python 3.10 or later. Verify by running: `python3 --version`
- Java 17 or later (required by Elasticsearch). Install via Homebrew: `brew install openjdk@17`
- Homebrew package manager. Install from <https://brew.sh> if not already present.
- Approximately 6 GB of free disk space and 4 GB of RAM.

2. Project Folder

- Download the project zip and extract it.
- Folder structure:

```
UniversityChatbot/
├── main.py
├── requirements.txt
├── utilities/
│   ├── slot_filling.py
│   ├── query_augmentation.py
│   ├── tuition_fee_kind.py
│   ├── es_client.py
│   ├── embedding_model.py
│   └── config/
│       └── vocabulary.json
├── data/
│   ├── calendar_events.json
│   ├── contacts_data.csv
│   ├── tuition_data.json
│   └── unstructured_chunks.json
├── indexing/
│   ├── calendar_index.py
│   └── contacts_index.py
```



- All commands in this guide run from inside the UniversityChatbot/ folder.

3. Python Environment Setup

- Open Terminal and navigate to the project folder:
`cd ~/UniversityChatbot`
- Create a virtual environment:
`python3 -m venv venv`
- Activate the virtual environment:
`source venv/bin/activate`

Note: Your terminal prompt will now show (venv) at the start. This prefix must be visible before running any project command. Everytime you open a new terminal window, re-run that activation line before using the project.

- Install dependencies:
`pip install -r requirements.txt`

This installs all required packages including sentence-transformers. The first time an indexing script runs, it downloads the E5-large-v2 embedding model (approximately 1.2 GB). This is automatic, requires an internet connection, and only happens once.

4. Setting up Elasticsearch 8.x

Elasticsearch is the search database the chatbot queries. Elasticsearch must be running before indexing or starting the chatbot.

- Downloading Elasticsearch:
 - Go to: <https://www.elastic.co/downloads/elasticsearch>
 - Select the macOS ([tar.gz](#)) version under the 8.x release and download it.
- Extracting Elasticsearch:
 - Open a terminal window, navigate to folder where the file is downloaded and run:

```
cd ~/<folder where Elasticsearch is downloaded>  
tar -xzf elasticsearch-8.x.x-darwin-aarch64.tar.gz
```
 - Replace 8.x.x with the actual version number and move to the extracted folder:

```
Cd elasticsearch-8.x.x
```

Note: This folder is separate from the UniversityChatbot/ project folder. The Elasticsearch folder can live anywhere on your machine.

4.1 Disabling security

Elasticsearch 8.x has security turned on by default, which requires HTTPS and password authentication. For local use, this must be turned off.

- From inside the elasticsearch-8.x.x/ folder, open the file config/elasticsearch.yml. The full path will be something like:

```
/Users/yourname/Downloads/elasticsearch-8.x.x/config/elasticsearch.yml
```

- Open it in any text editor (TextEdit, nano, VS Code). Find and set the following two lines. If they are not present, add them at the bottom of the file:

```
xpack.security.enabled: false  
xpack.security.http.ssl.enabled: false
```

- Save the file

4.2 Starting Elasticsearch

- From inside elasticsearch-8.x.x/:

```
./bin/elasticsearch
```

- Startup takes 20 to 30 seconds. Wait until a line containing **started** is seen in the output.
- Keep the terminal window open.

4.3 Verifying

- Open a new terminal window and run:
`curl http://localhost:9200`
- A block containing a "**cluster_name**" field should be seen, this confirms Elasticsearch is running.
- If you get a connection refused error, Elasticsearch has not finished starting. Wait a few more seconds and retry.

5. Configuring Environment Variables

- A .env.example file is included in the project folder. Copy it and rename the copy to .env:
`cp .env.example .env`
- Open .env in any text editor and fill in each value as described below.
 - a. ES_HOST
 - Leave this as <http://127.0.0.1:9200>.
 - This points to Elasticsearch running on your local machine.
 - b. GROQ_API_KEY
 - Go to <https://console.groq.com/keys>
 - Sign up for a free account.
 - Click "Create API Key", give it any name, and copy the key.
 - Paste it as the value for GROQ_API_KEY.
 - c. AZURE_OPENAI_ENDPOINT and AZURE_OPENAI_KEY
 - Go to <https://portal.azure.com> and sign in with your Microsoft account.
 - Search for "Azure OpenAI" in the top search bar and open the service.
 - Create a new resource if you do not have one. Select a region, give it a name, and complete the creation steps.

- Once the resource is created, go to "Keys and Endpoint" in the left sidebar.
 - Copy "Key 1" and paste it as `AZURE_OPENAI_KEY`.
 - Copy the "Endpoint" URL and paste it as `AZURE_OPENAI_ENDPOINT`.
 - Under "Model deployments", deploy a gpt-4o model. The deployment name you choose must match `AZURE_OPENAI_DEPLOYMENT` in the `.env` file.
- d. `AZURE_OPENAI_DEPLOYMENT`
- Set this to the deployment name you used when deploying the model in the step above. If you named it gpt-4o, leave the value as gpt-4o.
- e. `AZURE_OPENAI_API_VERSION`
- Leave this as 2025-01-01-preview. This is the API version the client uses when calling Azure OpenAI.
- Do not share the `.env` file or commit it to any public repository.

6. Loading and indexing files into Elasticsearch

- These four scripts create the Elasticsearch indices and load all university data.
- Before running them, confirm:
 - The (venv) prefix is visible in your terminal (if not, run `source venv/bin/activate`)
 - You are in the UniversityChatbot/ folder
 - Elasticsearch is running (Step 3 is complete)
- Calendar data (145 events):


```
python3 indexing/calendar_index.py
```
- Contacts data (1,552 records):


```
python3 indexing/contacts_index.py
```
- Tuition data (110 records):


```
python3 indexing/tuition_index.py
```
- Policy documents (599 chunks from 35 sources):


```
python3 indexing/documents_index.py
```
- Run them one at a time and wait for each to finish before starting the next. Each script prints progress as it runs. The first script will trigger the embedding model download and will take longer than the rest.
- If any script fails with a connection error, Elasticsearch is not running. Go back to Step 4

7. Start the API Backend

- Open a new terminal window. Navigate to the project folder and activate the virtual environment:

```
cd /UniversityChatbot
source venv/bin/activate
```
- Start the backend:

```
uvicorn main:app --reload --host 0.0.0.0 --port 8000
```
- To confirm it is running, open a browser and go to: <http://localhost:8000/health>
- The response will be:

```
{"status": "ok", "model": "chatbot_b"}
```
- If port 8000 is already in use:

```
lsof -i :8000
kill -9 <PID>
```

 - Then restart the unicorn.
- Keep this terminal window open.

8. Start the User Interface (Optional)

- Open a third terminal window. Navigate to the project folder and activate the virtual environment:

```
cd /UniversityChatbot
source venv/bin/activate
```
- This is to test if the UI is working properly, this is not the production UI rather a test UI for checking if the pipeline is working.

```
streamlit run core/pipeline.py
```

 - A browser window will open at <http://localhost:8501>.
 - The production API is at <http://localhost:8000>. Any external frontend connects to that endpoint directly

9. Test the system

- Type each of the following into the chat:
 - What are the fall 2025 tuition rates?
 - When is Thanksgiving break?
 - Contact information for Financial Aid
 - What is the academic probation policy?
- Each query should return a specific, grounded answer pulled from the indexed data.